Emiliano Chapa A01288300

Elliot Milan A01287593

Symptom Tracker Phase Two.

Project Description.

This is a Python terminal application that allows students to keep track of their symptoms, such as headaches, tiredness and lack of concentration, along with their daily habits, including sleep, meals, water intake and exercise. The main goal is to help them find simple patterns between their habits and how they feel each day.

Problem it solves.

Many people, especially students with busy schedules, experience symptoms regularly but do not always know what might be causing them. Since they don't keep track of their daily habits, it can be difficult to notice connections between things, like sleeping too little and getting a headache. This app helps them collect information and understand these patterns more easily.

The app does not provide medical diagnoses or replace a doctor. It only gives basic recommendations about daily habits.

How to Run the Code.

Install Python from the official Python website. Make sure Python is added to PATH during installation.

Open the project folder in Visual Studio Code and use its terminal to install the required libraries with the following command.

pip install -r requirements.txt

Run the program using the following command.

python main.py

To test the patterns without waiting several weeks to collect data, load the sample data using option five in the main menu.

Features Implemented.

The four features planned during Phase One now have a basic working version.

The daily tracking feature allows users to record their symptoms, intensity and daily habits. It is available through option one in the menu.

The history feature shows previous records in a table and includes a summary of average values. It is available through option two.

The pattern detection feature compares symptoms and daily habits to identify possible connections. It is available through option three.

The basic habit recommendations feature provides simple suggestions based on the collected information, without giving medical diagnoses. It is available through option four.

Option five loads simulated data, making it easier to test the application and see how it works.

How These Features Solve the Problem.

The daily tracking feature helps solve the lack of information. In less than a minute, users can record how they felt and what their day was like. This information is saved so they can review it later.

The history feature allows users to see their previous records and average values in one place. For example, they can check how many hours they usually sleep, which can be difficult to notice without keeping track of it.

The pattern detection feature compares daily habits with symptoms. For example, it might show that symptoms were reported on one hundred percent of the days when a person slept less than six hours, compared to ten percent of the days when they slept six hours or more. This helps users notice possible relationships between their habits and how they feel.

The recommendations feature turns these patterns into simple actions, such as trying to get more sleep or maintaining better daily habits. The purpose is to help users make more informed decisions. The app does not diagnose any medical condition and recommends consulting a healthcare professional if symptoms are severe or do not go away.

Programming Concepts Used.

Conditionals and loops are used to control the menu through while loops and if and elif statements. They also help validate user input and repeat questions until a valid answer is provided. For loops are used to go through the records and check different conditions.

Functions and modular code keep each task organized in its own function. The program is also divided into different modules, which makes the code easier to understand and maintain.

Data structures are used throughout the application. Dictionaries store records, symptom options and recommendations. Lists are used for conditions and results, while tuples contain the sample data.

Files are used to save and read the daily records from a CSV file. This means the information can still be available when the program is opened again.

An external library called pandas is used to read and write the CSV file. It also helps calculate averages, percentages and the most common symptom, making the analysis easier and more organized.

Project Structure.

The project folder is called rastreador fase2 and contains the main Python files and the data folder.

The file main.py contains the main program, including the menu and screens.

The file almacenamiento.py handles reading and writing the CSV file.

The file analisis.py is responsible for summaries and pattern detection using pandas.

The file recomendaciones.py contains the basic recommendations based on the detected patterns.

The file requirements.txt lists the libraries needed to run the application, including pandas.

The file README.md contains the project documentation and instructions.

The data folder contains the file registros.csv, which stores the daily records, with one row for each day.

CSV File Format.

The CSV file contains several columns that store the information collected by the application.

The fecha column stores the date of the record in year, month and day format.

The sintoma column stores the symptom reported by the user, such as no symptoms, headache, tiredness, lack of concentration or another symptom.

The intensidad column stores the symptom intensity. Zero means there were no symptoms, while values from one to five represent the intensity, from mild to very strong.

The sueño column stores the number of hours the user slept.

The comidas column records the number of meals eaten during the day.

The agua column records the number of glasses of water consumed.

The ejercicio column indicates whether the user exercised that day.

The ejemplo column indicates whether the record comes from sample data or was entered by the user.

If a user enters information twice for the same date, the existing record is updated instead of creating a duplicate.

Next Steps for the Final Phase.

For the final phase, the plan is to add a graphical interface with a window and buttons, making the application easier to use.

Another improvement will be the option to filter the history by symptom, so users can find specific records more quickly.

Finally, the application will include graphs to display the detected patterns in a more visual way. This should make the information easier to understand and help users identify possible connections between their daily habits and symptoms.